

Image Indexing and Retrieval

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Lecture Overview: Image Indexing and Retrieval

- ▶ Retrieval approaches: metadata, text annotation, content-based (CBIR)
- ▶ Color features: histograms, distances, spatial layout, color spaces
- ▶ Shape features: segmentation, grid encoding, invariance
- ▶ Texture, composition, and compressed-domain retrieval
- ▶ Goal: index images by visual content and enable query-by-example in a multimedia DB



Image Indexing and Retrieval

► Main Techniques

1. Based on the general properties of the image and stored as tables and queries using a relational data model.
 - Example: photo name, type of image, date of creation, subject, author,...
 - Disadvantage: difficult to store the true content of the photo
2. Based on a combination of extracting the features of the image with the recognition of objects in the image. Cons: complex computation.
3. Use text to describe photos, then apply IR systems to store and query photos
4. Use low-level attributes (color, context) for image indexing and retrieval

The 2-4 techniques are called content-based image data queries.

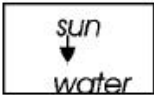
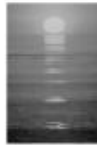
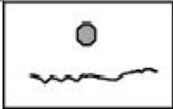
Content-Based Image Retrieval (CBIR) Framework

- ▶ Offline: image → preprocess → segment (optional) → extract features → index → MMDB
- ▶ Online: query image or sketch → extract features → similarity search → ranked results
- ▶ Four main approaches:
 - ▶ 1. Metadata (relational): title, author, date — fast but no content search
 - ▶ 2. Feature + object recognition: high accuracy, high compute
 - ▶ 3. Text annotation + IR: semantic keywords (Lecture 8 inverted index)
 - ▶ 4. Low-level CBIR: color, shape, texture — query-by-example without text
- ▶ Semantic gap: low-level features $\neq$ user concepts ('sunset', 'happy person')



Image low-level attributes

- ▶ Colors, shapes of objects, background colors
- ▶ Parameters that present object dimensions
 - ▶ Example: Rectangle with $XLB = 80$; $XUB = 120$; $YLB = 20$; $YUB = 55$.
- ▶ Parameters that present the color values of a pixel
 - ▶ Example: at a pixel with coordinates $(200, 150)$ there are color values as Red = 12; Green = 7; Blue = 8.

	Example query	Example query result
exact	Spatial predicate 	  
	Image predicate <i>Amount of "sky" > 20% and amount of "sand" > 30%</i>	  
	Group predicate <i>Location = "Africa"</i>	  
approximate	Spatial example 	   
	Image example 	   
	Group example <i>pos</i>  <i>neg</i>  	   

Comparison of Image Retrieval Approaches

- ▶ Metadata: exact filter queries; no similarity; requires manual cataloging
- ▶ Text IR: rich semantics; manual/auto caption; poor for 'find similar image'
- ▶ CBIR color: fast index; good for color/layout; weak on semantic objects
- ▶ CBIR shape: good for logos, parts; needs segmentation; rotation/scale handling
- ▶ Hybrid (best practice): metadata filter → CBIR re-rank → text expand
- ▶ MMDB stores: image OID + feature vectors + text index + EXIF metadata

Text-based image data retrieval

- ▶ Photos are depicted in free-form text
- ▶ Queries consist of keywords or free text with or without boolean operators.
- ▶ Query results are based on a comparison of similarities between the query and the archived content (IR systems)

Text-based image data retrieval

- ▶ The text used to describe the photo must be input manually (or from built-in captions)
 - ▶ How to enter photo descriptions accurately and consistently
 - ▶ Need to use contextual information, or multiple synonyms to enrich the information about the photo
 - ▶ The combination of different words also carries a lot of different information
- ▶ The text sometimes doesn't cover the full meaning of the photo, or sometimes it's too specific.
 - ▶ The query process requires a feedback.

Text-based image data retrieval

- ▶ Capable of containing semantic information (higher-level attributes) in an accurate and detailed manner. For example: mood information such as “smiling”, “happiness”...
- ▶ Cons: Some low-level attributes are difficult to express in text, such as photo backgrounds, or specially shaped objects.
- ▶ Text queries are difficult to apply in live image queries. For example: “Find images similar to this one...”

Text-Based vs Content-Based Image Retrieval

- ▶ Text strengths: 'smiling', 'happiness', event names — high-level semantics
- ▶ Text weaknesses: background color, exact shape, subtle texture hard to describe
- ▶ CBIR strengths: query-by-example — submit image, get similar images
- ▶ CBIR weaknesses: cannot directly query 'sad mood' without labeled training data
- ▶ Relevance feedback: user marks relevant/irrelevant results → refine query vector
- ▶ Modern systems: CLIP/CNN embeddings bridge text and image in shared vector space



Image retrieval based on color

▶ Basic Techniques

- ▶ Each image in the database is represented by basic colors (example: RGB)
- ▶ Each basic color band is subdivided into m levels. In total, there are $n = m^3$ color levels corresponding to the number of color cells
- ▶ Construct a color histogram $H(M) = [h_1, h_2, \dots, h_j, \dots, h_n]$ where h_j is the number of pixels in color-level j .
- ▶ $H(M)$ is stored and queried using vectorspace-based, or cluster-based query techniques.

Color Histogram — Construction and Indexing

- ▶ Quantize each RGB channel into m levels $\rightarrow m^3$ bins (e.g., $m=4 \rightarrow 64$ bins)
- ▶ Count pixels per bin $\rightarrow$ histogram $H = [h_1, h_2, \dots, h_n]$ where $\sum h_j = N$ (total pixels)
- ▶ Normalize: $h_j / N \rightarrow$ probability distribution (scale-invariant to image size)
- ▶ Index structure: vector database or inverted file on dominant color bins
- ▶ Query: compute $H_{\text{query}} \rightarrow$ rank images by distance/similarity to H_{query}
- ▶ Fast but discards spatial layout — two images with same colors in different positions match



Image retrieval based on color

- ▶ Measures of similarity, or measures of distance (L1-norm, L2-norm) between color histograms are used.
- ▶ For example, there are 3 8x8-point grayscale photos, let's say we use 3 bits to encode the grayscales:

$$H_1 = [8, 8, 8, 8, 8, 8, 8, 8]$$

$$H_2 = [6, 6, 8, 8, 10, 10, 7, 9]$$

$$H_3 = [10, 10, 2, 2, 10, 10, 10, 10]$$

Which two photos are the most similar according to the general similarity formula, or the L2-norm distance formula?

Histogram Distance Measures

- ▶ L1 (Manhattan): $d_{L1} = \sum |H_1(i) - H_2(i)|$ — sum of absolute bin differences
- ▶ L2 (Euclidean): $d_{L2} = \sqrt{\sum (H_1(i) - H_2(i))^2}$ — most common in CBIR
- ▶ Histogram intersection: $\text{sim} = \sum \min(H_1(i), H_2(i))$ — higher = more similar
- ▶ Chi-square: $\chi^2 = \sum \frac{(H_1(i) - H_2(i))^2}{H_1(i) + H_2(i)}$, sensitive to small bins
- ▶ Bhattacharyya: $\text{sim} = \sum \sqrt{H_1(i) \times H_2(i)}$ — geometric mean overlap
- ▶ Choose L2 for equal-sized normalized histograms; intersection for unnormalized counts



Image retrieval based on color

- ▶ Example of color histogram:

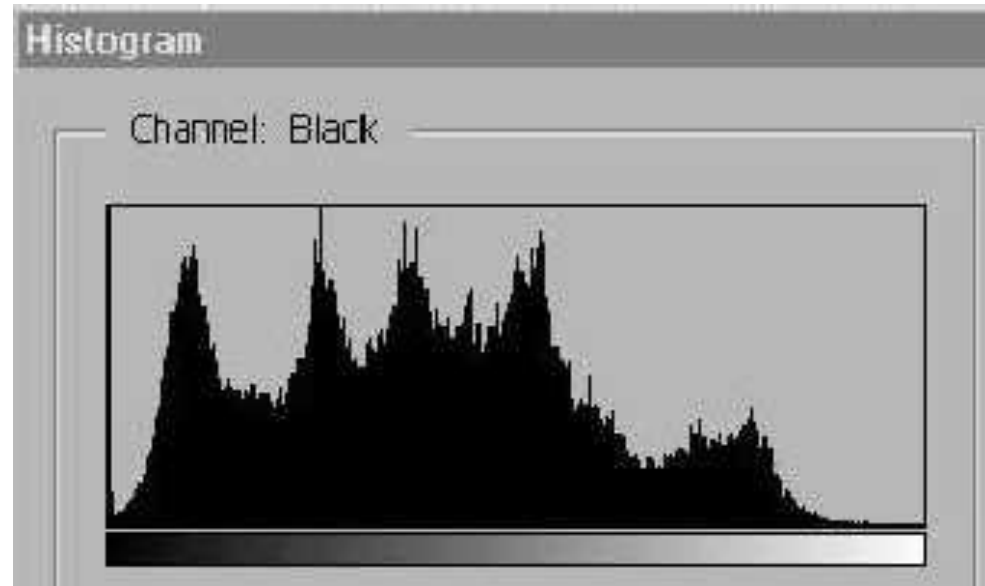
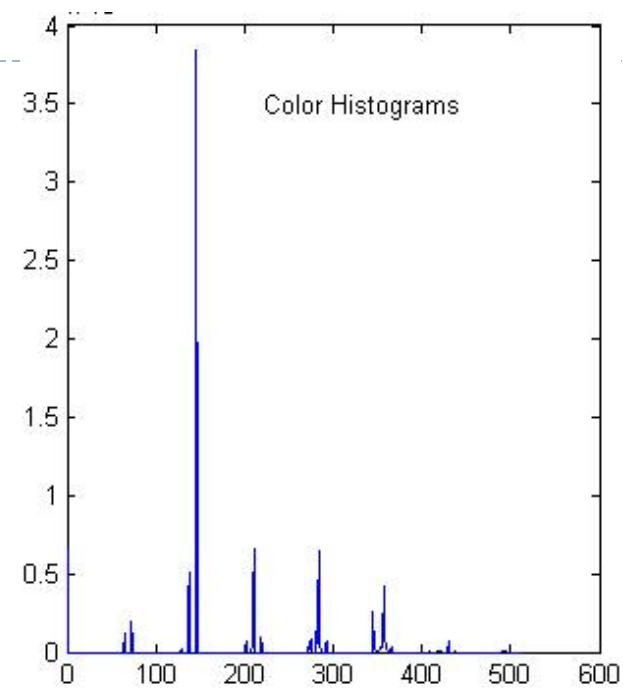
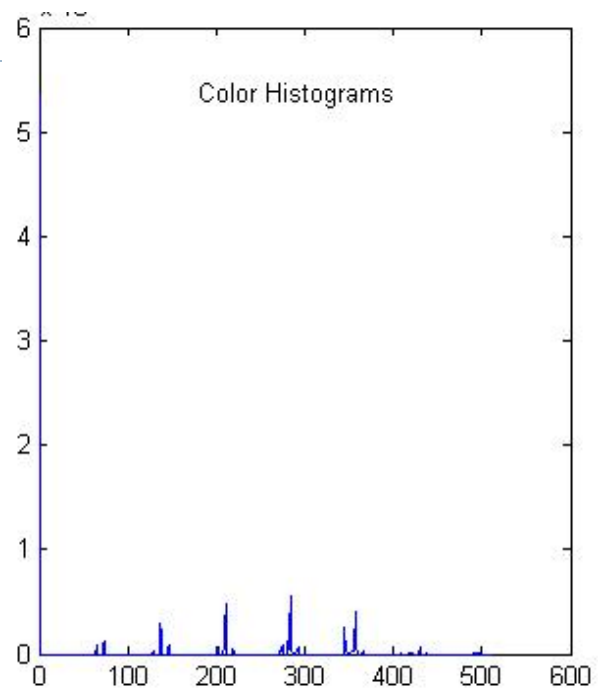


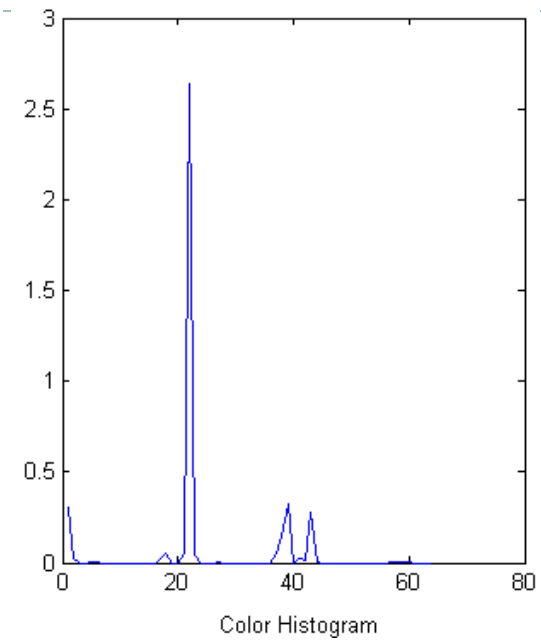
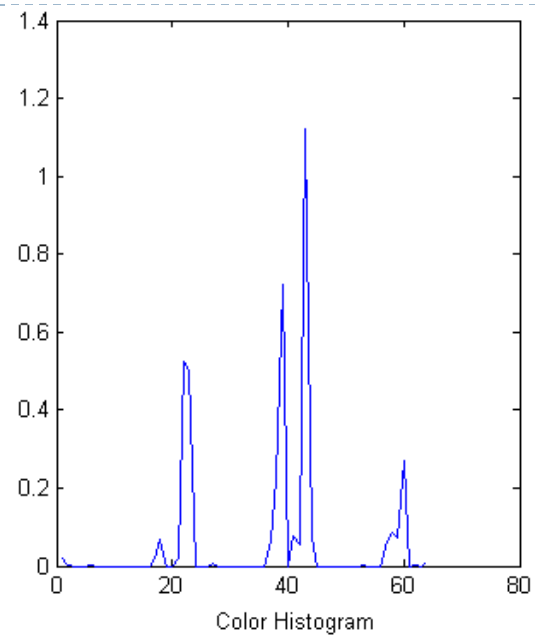
Image retrieval based on color

- ▶ Advanced techniques for color histograms
 - ▶ Consider the color similarity:
 - ▶ Assuming each image consists of N pixels, if L1-norm is used, the maximum distance between the 2 images will be $2N$, which occurs when the two images do not have any color value in common.
 - ▶ In fact, the human eye still perceives color similarities even though their color values are different..
 - ▶ Sometimes the color value is changed due to noise and fluctuations in brightness,... resulting in two similar colors that can belong to two different color levels. Or sometimes in the process of querying, we need to find images with similar colors, not necessarily with equal color values.



Perceptual Color Similarity and Weighted Histograms

- ▶ Problem: RGB bin boundaries treat (255,0,0) and (254,0,0) as different bins
- ▶ Cross-bin similarity matrix A : $a_{ij} = 1 - \frac{d(\text{color}_i, \text{color}_j)}{d_{\max}}$
- ▶ Weighted distance: $\|Z\|^2 = Z^T A Z$ where $Z = H_X - H_Y$ (perceptual penalty)
- ▶ Perceptually Weighted Histogram (PWH): weight each pixel by inverse distance to rep. colors
- ▶ Cumulative histogram: prefix sum of sorted bins — coarse but fast; loses detail
- ▶ Better color spaces: CIE Lab — distance approximates human perceptual difference (ΔE)



Spatial Color Layout — Block Histograms

- ▶ Divide image into $k \times k$ grid (e.g., $3 \times 3 = 9$ regions)
- ▶ Compute separate color histogram H_r for each region r
- ▶ Feature vector: concatenate $[H_1, H_2, \dots, H_{k^2}]$ — captures rough spatial layout
- ▶ Query: compare region r of query with region r of database images (or allow shift tolerance)
- ▶ Masking effect: large uniform background dominates small object regions
- ▶ Mitigation: separate foreground/background histograms; weight object regions higher

Image retrieval based on color

- ▶ Use a Cumulative Histogram $CH(M) = [ch_1, ch_2, \dots, ch_n]$ where $ch_i = \sum_{j \leq i} h_j$
 - disadvantage: Doesn't accurately express the similarity between colors
- ▶ Change the color similarity calculation formula
 - Let X and Y are the color histograms of the 2 images and Z is the vector difference between X and Y. The similarity between X and Y is calculated as $||Z|| = Z^T A Z$, where: $a_{i,j} = 1 - d(c_i, c_j)/d_{\max}$, is the color gap in the color space.
- ▶ Use Perceptually Weighted Histogram (PWH)
 - Select N representative colors corresponding to N frequency chart cells.
 - Calculate the distance between the color of each pixel and those representative colors. The inverse of each distance is the weight of each pixel to each color representation.
 - All weights of each representative color will be added to each bin.

Image retrieval based on color

- ▶ The disadvantage of color-based image querying is that it does not take into account the spatial relationship between pixels.
- ▶ Consider spatial relationship between pixels
 - ▶ The photo is divided into many small fixed areas
 - ▶ At each image area, a color histogram is constructed
 - ▶ During the query process, the color histogram in the corresponding regions will be compared with each other.

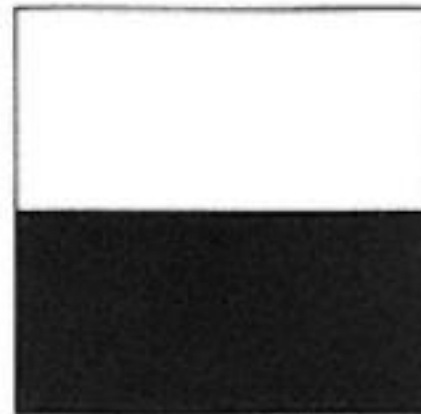


Image retrieval based on color

- ▶ The disadvantage of this method is the influence of the masking effect.
 - ❑ Large colored tiles will overwhelm the contents of small colored tiles.
 - ❑ Photos with the same background color tend to have the same content, even though the objects/content in them are completely different.
- ▶ Separate color histograms can be used for backgrounds and objects: divide the image into multiple regions, each containing its own object or background image.
- ▶ The identification of the object or background image is implemented based on the variation of the color value along the horizontal or vertical axis
- ▶ Each photo is represented by multiple color histograms.
- ▶ The query process can be done in one of the following ways:
 - ❑ Based on the overall color histogram of the image
 - ❑ Based on color histograms of objects in the image
 - ❑ Based on the color histogram of the background
 - ❑ Based on the color histograms of both the backgrounds and the objects (with different weights)

Image retrieval based on color

- ▶ Using statistical information about the color distribution (like nonlinear quantization): the color range with a more pixel distribution will be subdivided.
- ▶ Use different color expressions
 - ▶ The RGB system has some disadvantages:
 - Depends on the color formatting device (color matching principle)
 - Inconsistency in the similar color representation.
 - ▶ Change the color space from RGB to CIE Luv, CIE Lab, HSI (hue, saturation, intensity), HVC (hue, value, chroma)
 - ▶ Change the image expression: different photography devices have different adjustment parameters for the photos. The same image captured from different sources may have different color parameters

Color Spaces for CBIR

- ▶ RGB: device-dependent; not perceptually uniform; sensitive to lighting changes
- ▶ HSI/HSV: separate hue (color type), saturation (purity), intensity — robust to shadows
- ▶ CIE Lab: designed for perceptual uniformity; $\Delta E < 2$ = imperceptible difference
- ▶ CIE Luv: alternative perceptual space used in MPEG-7 ColorLayout descriptor
- ▶ Practice: index in HSV or Lab; store conversion parameters in metadata
- ▶ Illumination normalization: divide by average intensity before histogram computation



Image retrieval based on shapes

- ▶ An image needs to be divided into segments containing objects
 - ▶ Each object in the image needs to be expressed systematically, regardless of the variation of scales, positions, and sizes,...
 - ▶ Similar objects need to be expressed the same way
- ▶ During the query process, the user uses a sample image or create a pattern himself so that the system searches for images with similar objects.
- ▶ The results usually include multiple photos, arranged in descending order of similarity to the sample image.

Shape-Based Retrieval Pipeline

- ▶ 1. Segmentation: separate object from background (threshold, GrabCut, edge detection)
- ▶ 2. Normalization: translate to centroid, scale to unit size, rotate to align main axis
- ▶ 3. Feature extraction: grid binary, contour, Hu moments, Fourier descriptors
- ▶ 4. Index: store feature vector linked to image OID + bounding box metadata
- ▶ 5. Query: user submits sketch or sample image → same pipeline → k-NN search
- ▶ Challenge: same object at different scales/rotations must map to similar features



Image retrieval based on shapes

- ▶ General concepts and some measures of basic shapes.
 - ▶ Main axis: a line connecting the two farthest points in the figure
 - ▶ Auxiliary axis: a line perpendicular to the main axis with a length such that a rectangle with parallel sides and same length with the main and the auxiliary axes captures the entire envelope of the figure.
 - ▶ Basic rectangle: formed as described above
 - ▶ Eccentricity: the ratio of the length between the main and the auxiliary axes
- ▶ The above basic information can be used to query images based on shapes.

Image retrieval based on shapes

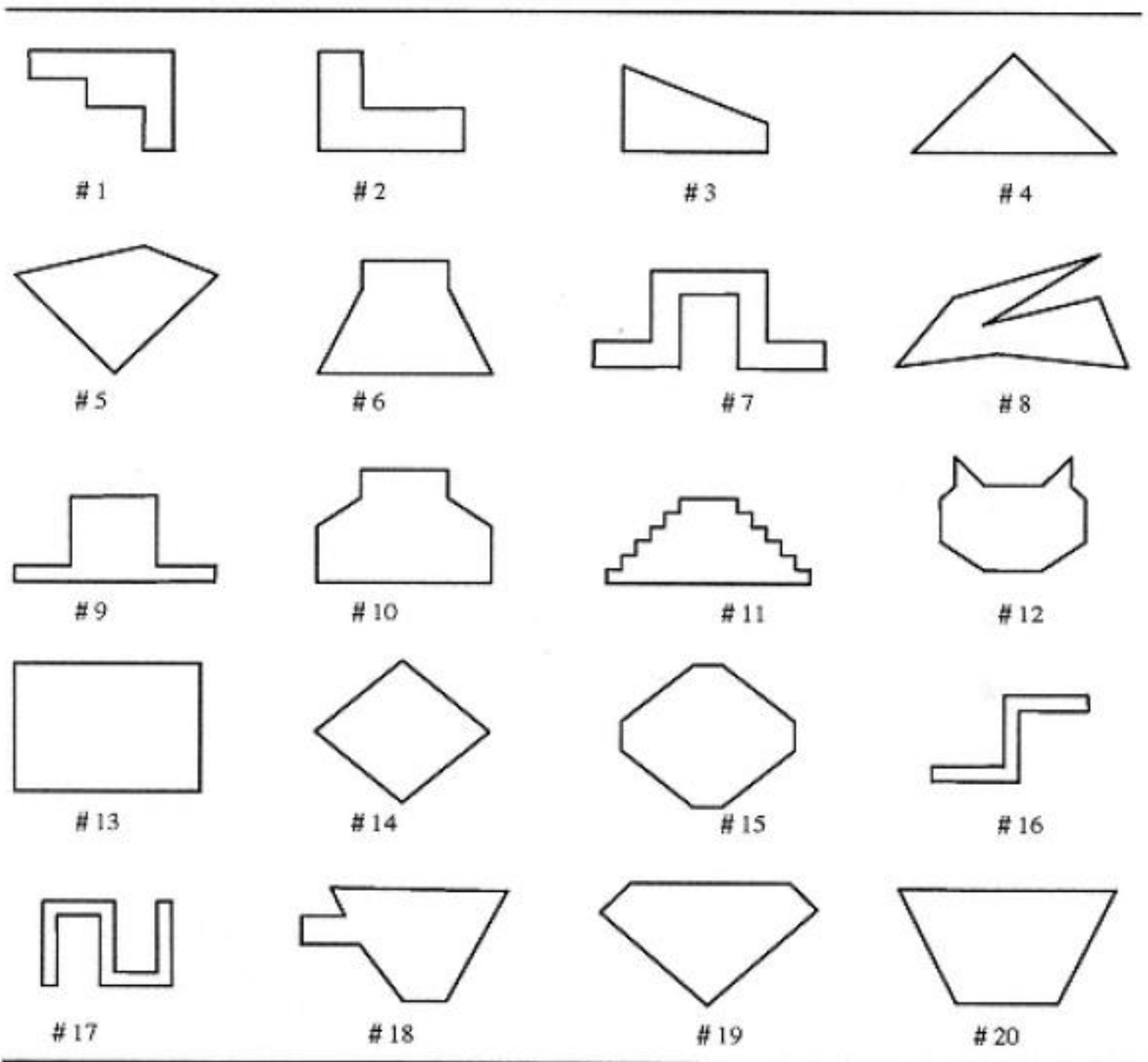
- ▶ Describe shapes by regions and similarity comparison criteria
 - ▶ Basics of regional shape representation:
 - ▶ For each shape, we place a grid (consisting of squares of a fixed size) on it
 - ▶ For each square with more than 15% of the area of the object, we assign a value of 1
 - ▶ The remaining shapes have a value of 0.
 - ▶ Read the value of the squares from left to right, top to bottom to have a binary sequence that represents that shape.
 - ▶ Rotation normalization
 - ▶ Length normalization: all main axis of the same length (192)
 - ▶ Unified shape expression – shape indexing
 - ▶ Measures of similarity.
 - ▶ Other techniques based on shapes

Advanced Shape Descriptors

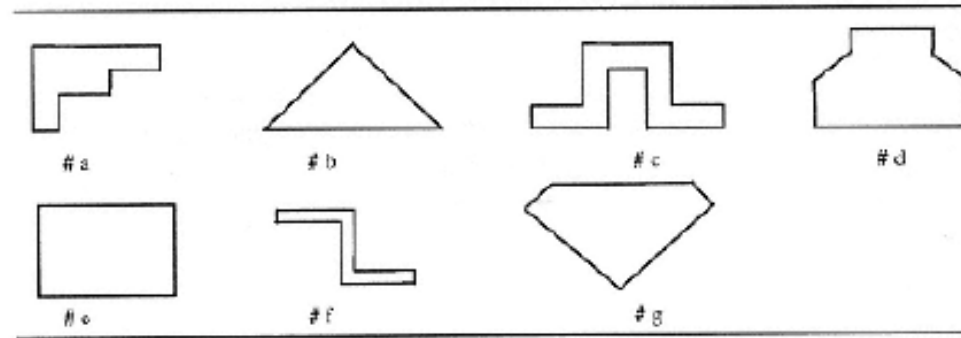
- ▶ Hu invariant moments: 7 values invariant to translation, scale, rotation
- ▶ Fourier descriptors: FFT of boundary contour → magnitude invariant to rotation
- ▶ Chain code: encode boundary direction (4-direction or 8-direction) as symbol string
- ▶ SIFT/SURF keypoints: scale-invariant feature points with orientation (object-level CBIR)
- ▶ Deep CNN embeddings (ResNet, VGG): state-of-the-art general CBIR feature vectors
- ▶ Trade-off: moments/Fourier fast but coarse; CNN slow to compute but semantically rich



Database of shapes



Query shapes



Shape Database Indexing

- ▶ Each indexed shape entry stores:
 - ▶ shape_id, source_image_oid, bounding_box (x_{lb} , y_{lb} , x_{ub} , y_{ub})
 - ▶ binary grid vector (192-bit after length normalization) or moment vector
 - ▶ eccentricity, main_axis_angle, area, perimeter
- ▶ Index structures: sequential scan (small DB); R-tree on bounding boxes; VP-tree on Hamming
- ▶ Multi-resolution: store coarse grid (8×8) for filter + fine grid (32×32) for re-rank
- ▶ Link to MMDB object table: one image may contain multiple shape index entries

Query-by-Example Shape Retrieval

- ▶ User query types: (1) sample image patch, (2) hand-drawn sketch, (3) template icon
- ▶ System pipeline: query → segment → normalize → extract features → top-k retrieval
- ▶ Results ranked by ascending distance (Hamming, L2 on moments, chamfer on contours)
- ▶ Relevance feedback: user selects best match → update query vector toward selected shape
- ▶ Applications: trademark/logo search, part identification, CAD component retrieval
- ▶ Combine with color filter first: 'red objects shaped like circle' → color pre-filter + shape rank

Distance Measures for Image Features

- ▶ Feature vectors (color histograms): L1, L2, cosine distance, χ^2 , EMD (Earth Mover's Distance)
- ▶ Binary shape grids: Hamming distance = number of differing bits
- ▶ Cosine similarity: $\cos(\theta) = (A \cdot B) / (|A||B|)$ — good for high-dimensional sparse histograms
- ▶ EMD: minimum cost to transform one histogram into another — handles partial overlap
- ▶ MPEG-7: standardizes ColorLayout, EdgeHistogram, ShapeBlob descriptors + distances
- ▶ Choice depends on feature type, dimensionality, and need for perceptual weighting

Distance measures

	a	b	c	d	e	f	g
1	0	104	46	76	37	44	65
2	33	65	69	80	38	61	56
3	17	81	33	96	48	37	60
4	104	0	87	76	71	100	37
5	72	62	75	41	53	78	61
6	98	55	104	50	110	103	77
7	46	87	0	86	79	44	60
8	54	88	32	117	89	40	69
9	32	84	34	101	55	52	65
10	76	76	86	0	72	93	76
11	40	76	50	89	47	64	59
12	105	68	121	65	69	90	80
13	37	71	79	72	0	71	66
14	108	55	116	64	74	91	69
15	123	71	132	62	88	137	62
16	44	100	44	93	71	0	63
17	54	92	71	109	49	50	91
18	54	51	52	54	62	69	36
19	65	37	60	76	66	63	0
20	91	38	109	65	81	86	70

Texture, Moments, and Border Features

- ▶ Invariant moments: capture global shape distribution without explicit boundary
- ▶ Fourier coefficients: periodic texture patterns (fabric, brick walls)
- ▶ Border color histogram: colors along object contour — captures outline appearance
- ▶ Focal pixel directory: salient points with color/gradient — compact summary
- ▶ Flexible pattern matching: elastic template matching tolerates local deformation
- ▶ GLCM (Gray-Level Co-occurrence Matrix): texture contrast, homogeneity, energy



Image retrieval

- ▶ Invariant moments

$$m_{pq} = \sum_x \sum_y x^p y^q f(x, y)$$

- ▶ Fourier coefficients
- ▶ Color histogram of main borders
- ▶ Directory of focal pixels
- ▶ Comparison of Flexible Patterns

Image retrievals based on composition

- ▶ Attributes that are mainly related to image composition
 - ▶ Roughness of an image (as opposed to smoothness): the difference between elements of an image
 - ▶ Contrast: the dynamic range of the gray band, the polarization between light and dark colors in the histogram, the sharpness of the strokes, the frequency of repetition of the color patterns
 - ▶ The shape and direction of the objects in the photo
 - ▶ Boundary similarity: compare edge maps or contour orientation histograms
 - ▶ Color composition order: horizontal/vertical color band sequences (sunset strip pattern)
 - ▶ Useful for: art retrieval, wallpaper search, aesthetic classification
 - ▶ Often combined with color and texture into a single extended feature vector

Compressed image retrieval

- ▶ Indexing and retrieval based on DCT coefficients
- ▶ Indexing and retrieval based on FFT coefficients
- ▶ Indexing and retrieval based on Wavelet coefficients
- ▶ Indexing and retrieval based on compressed VQ

Compressed-Domain Image Retrieval

- ▶ Idea: search in JPEG/MPEG bitstream without full decompression — faster for large archives
- ▶ DCT coefficients: DC values approximate block average color; AC patterns capture texture
- ▶ FFT/Wavelet coefficients: frequency-domain energy distribution as feature vector
- ▶ VQ (Vector Quantization) codebook indices: cluster of DCT blocks as compact descriptor
- ▶ Advantage: reuse compression artifacts as features; no decode overhead
- ▶ Limitation: features tied to codec parameters; not portable across re-encoded images



Summary — Image Indexing and Retrieval

- ▶ Four retrieval paradigms: metadata, text IR, CBIR features, hybrid
- ▶ Color histograms: fast, effective; extend with spatial layout and perceptual color spaces
- ▶ Shape features: require segmentation and invariance normalization
- ▶ Texture/composition add discriminative power for structured and artistic images
- ▶ Compressed-domain retrieval avoids decode cost for JPEG archives

